GAME THEORY:

APPLIED TO COMPETITON

Session 16

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1. Mixed strategies

2. Applying game theory to a “real” strategy case:

   Deregulation of Spain’s passenger rail industry.

Mixed strategies

Game theory when there is no single, clear equilibrium

- **Pure Strategies** where we assume that all the strategies are chosen with the same probability.

Person 1 \ Person 2	Apply	Don't apply
Apply	(1, 4)	(3, 2)
Don't apply	(2, 3)	(1, 2)

ONE
Nash Equilibrium

✓ **Unique pure-strategy NE**

(Don't apply, Apply) with payoffs (2, 3)

Neither player can improve their payoff by changing their action alone. *Each player chooses one action with certainty*

What if there is no single, clear pure-strategy equilibrium?

1. There may be more than one pure-strategy Nash equilibrium

2. There may be NO pure-strategy equilibrium at all

Nash Equilibrium (NE) in **Mixed Strategies** is computed when each player assign a probability to each strategy.

1 There may be more than one pure-strategy Nash equilibrium

- A couple wants to spend time together.
- They can attend either the opera or a soccer match, but cannot communicate.
- Both prefer being together, but they prefer different activities
- Are there any pure-strategy equilibria?

	Opera	Soccer
Opera	(2, 3)	(0, 0)
Soccer	(0, 0)	(3, 2)

Two pure-strategy Nash equilibria, but not clear which one the players will select.

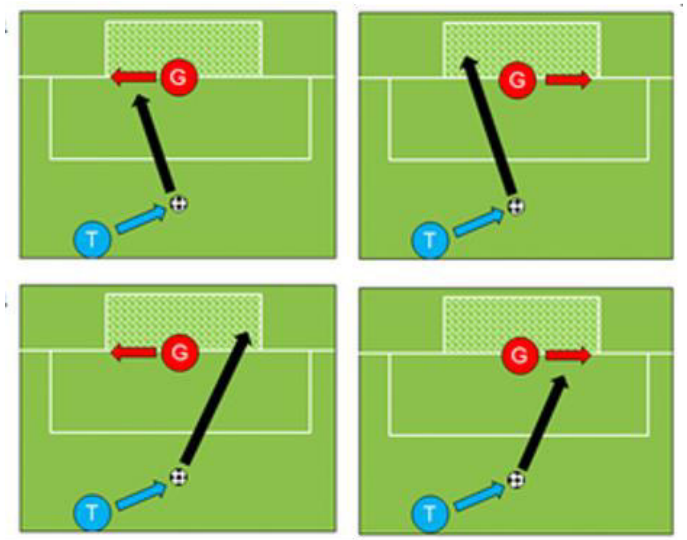
2 **There may be NO pure-strategy equilibrium at all**

Soccer penalty game

NO pure strategy equilibrium

Goalie

Shooter	Left	Right
Left	(4, 1)	(1, 5)
Right	(1, 2)	(2, 1)



- **Shooter** shoots left or right (from his/her perspective)
- **Goalie** must anticipate and dive left or right (from his/her view) — the goalie cannot wait for the shot and then dive

? **Are there any pure-strategy Nash equilibria?**
No. Every outcome gives one player an incentive to change

- When there is no unique pure-strategy equilibrium, we can look for a mixed-strategy Nash equilibrium.
- A pure strategy selects one action with certainty. *Example: always shoot Left.*
- A mixed strategy assigns a probability to each available action
  - *Example: shoot Left 20% and Right 80%. Probabilities do not need to be equal.*
- Players randomize when being predictable would allow the opponent to exploit them.

Tenis Player 2

Tenis Player 1	Left	Right
Left	(0, 2)	(6, 0)
Right	(8, 0)	(0, 4)

What happen if a tennis player always pays right or left?

Optional: https://www.youtube.com/watch?v=-RiZiTz2hWA



- To calculate the equilibrium:
  - Find the probabilities that make the opponent indifferent between their actions
  - Use each player's payoffs to calculate the other player's probabilities
- In a mixed-strategy Nash equilibrium, neither player can improve their expected payoff by changing their own probabilities alone.

Recap: Mixed strategies

When there is no unique pure-strategy equilibrium, we can look for a mixed-strategy Nash equilibrium

Pure strategy

Selects one action with certainty

Example: always shoot Left

Mixed strategy

Assigns a probability to each available action — the probabilities do not need to be equal

Example: shoot Left 20% and Right 80%

! Why randomize?

Players randomize when being predictable would allow the opponent to exploit them

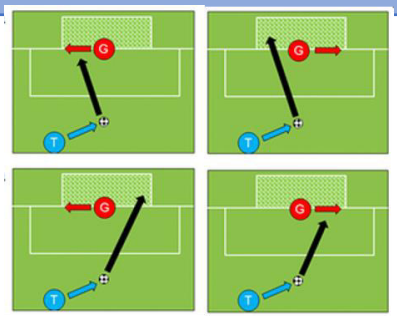
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J.Glauber – 2026 – Strategy

Soccer penalty game



Goalie

Shooter	**Left** P(Left) = q	**Right** P(Right) = 1 - q
Left P(Left) = p	(4, 1)	(1, 5)
Right P(Right) = 1 - p	(1, 2)	(2, 1)

Shooter's expected payoff from Left: 4q + 1(1 - q)

Shooter's expected payoff from Right: 1q + 2(1 - q)

Goalie's expected payoff from Left: 1p + 2(1 - p)

Goalie's expected payoff from Right: 5p + 1(1 - p)

1p + 2(1 - p) = 5p + 1(1 - p)
1p + 2 - 2p = 5p + 1 - 1p
-1p + 2 = 4p + 1
-5p = -1
p = 0.2 (20% shooter chooses left)

4q + 1(1 - q) = 1q + 2(1 - q)
4q + 1 - 1q = 1q + 2 - 2q
3q + 1 = -1q + 2
4q = 1
q = 0.25 (25% goalie chooses left)

Interpretation

- Shooter: Left 20%, Right 80% / Goalie: Left 25%, Right 75%
  - Given the goalie's probabilities:
    - $EU_{Shooter}(L) = 4(0.25) + 1(0.75) = 1.75$
    - $EU_{Shooter}(R) = 1(0.25) + 2(0.75) = 1.75$
  - Given the shooter's probabilities:
    - $EU_{Goalie}(L) = 1(0.20) + 2(0.80) = 1.80$
    - $EU_{Goalie}(R) = 5(0.20) + 1(0.80) = 1.80$

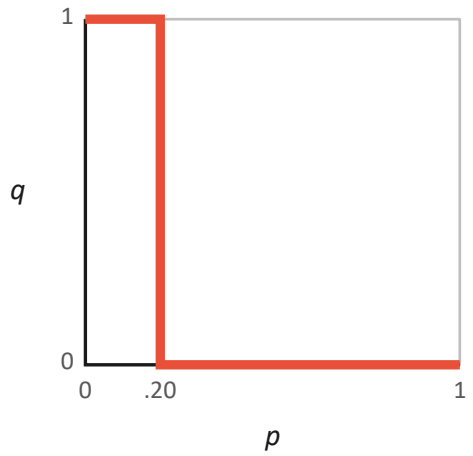
Substitute the opponent's equilibrium probabilities into the expected-payoff equations

- **Each player is indifferent between their two actions: both actions produce the same expected payoff. Therefore, neither player can improve their expected payoff by changing strategy alone**
- The equilibrium doesn't predict the outcome of one shot, but rather the probabilities with which players choose their actions across comparable situations.

Best response functions

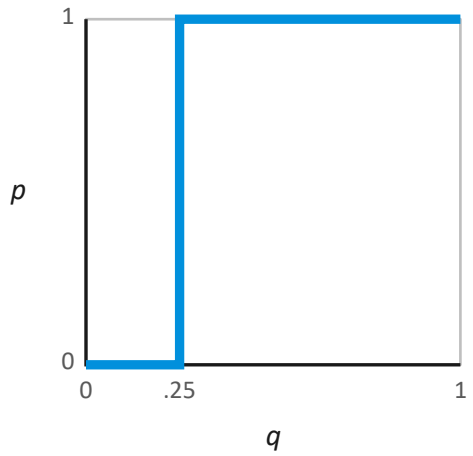
Goalie's best response

Indifferent when p = .20



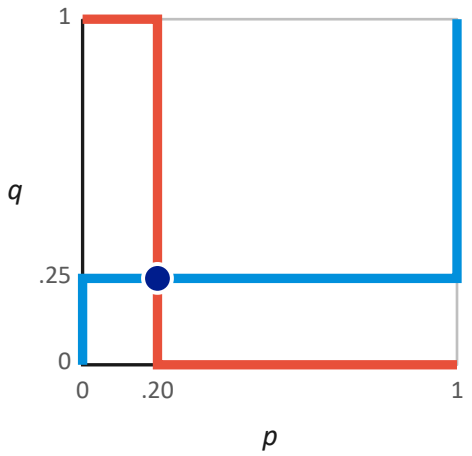
Shooter's best response

Indifferent when q = .25



Mixed-strategy NE

Occurs at p = .20, q = .25



J.Glauber – 2026 – Strategy

The equilibrium is where each player's strategy is a best response to the other's

Case: Netflix vs. Disney+

- Netflix and Disney+ must choose when to release major new content. Their expected profits depend on both platforms' release dates. The companies choose simultaneously, and neither wants its schedule to become predictable.
- What's the equilibrium?

NO pure strategy equilibrium

NETFLIX

Disney+

	Friday release	**Sunday release**
Friday release	($50, $50)	($80, $20)
Sunday release	($90, $10)	($20, $80)

*Payoffs represent expected profits in $ millions



P(Friday) = q P(Friday) = 1-q

P(Friday Netflix) = p



P(Friday Netflix) = 1-p

	Friday release	**Sunday release**	
Friday release	($50, $50)	($80, $20)	50q + 80(1 − q)
Sunday release	($90, $10)	($20, $80)	90q + 20(1 − q)
	50p + 10(1 − p)	20p + 80(1 − p)	

50p + 10(1 − p) = 20p + 80(1 − p)
50p + 10 − 10p = 20p + 80 − 80p
40p + 10 = 80 − 60p
100p = 70
p = 70/100 = 0.70

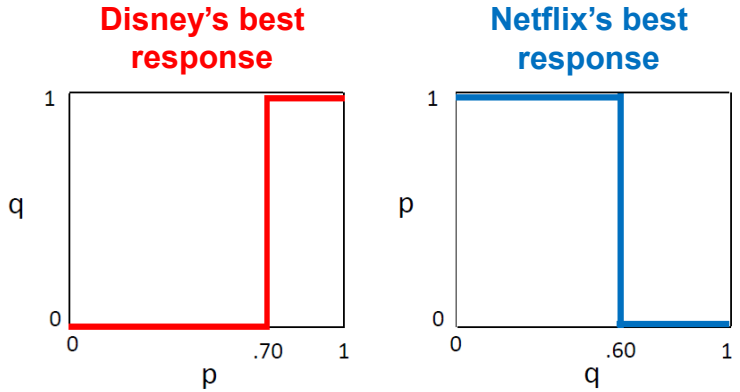
Netflix chooses Friday with prob 70%

50q + 80(1 − q) = 90q + 20(1 − q)
50q + 80 − 80q = 90q + 20 − 20q
80 − 30q = 70q + 20
60 = 100q
q = 60/100 = 0.60

Disney+ chooses Friday with prob 60%

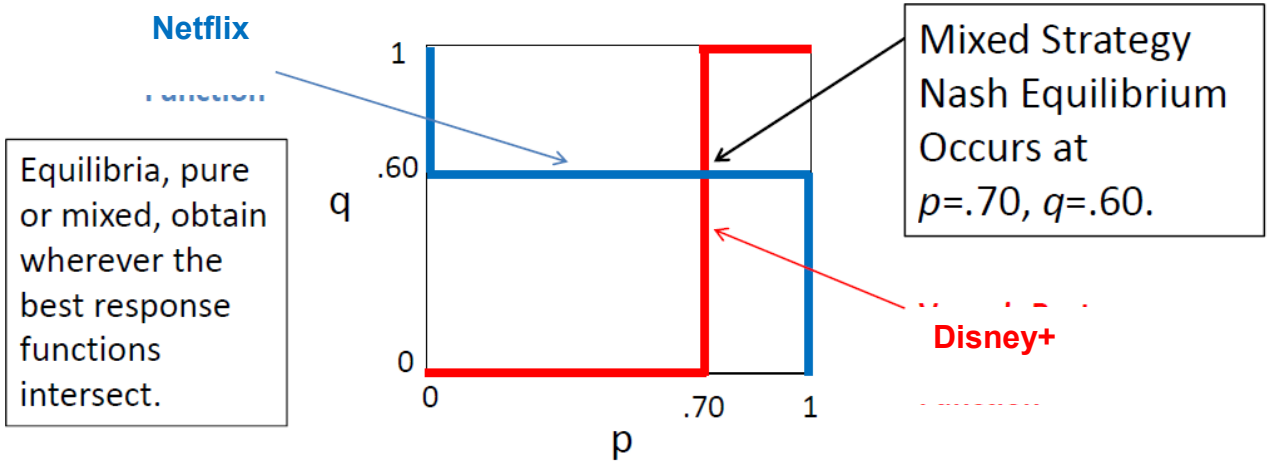
Interpretation

- Netflix: Friday release 70%, Sunday release 30%
- Disney+: Friday release 60%, Sunday release 40%
- Given Disney+'s probabilities:
  - $EU_{\text{Netflix}}(F)$ = 50(.60) + 80(.40) = 62
  - $EU_{\text{Netflix}}(S)$ = 90(.60) + 20(.40) = 62
- Given Netflix's probabilities:
  - $EU_{\text{Disney}}(F)$ = 50(.70) + 10(.30) = 38
  - $EU_{\text{Disney}}(S)$ = 20(.70) + 80(.30) = 38
- …both firms are indifferent between Friday and Sunday. Neither can improve its expected payoff by changing its probabilities alone



Combined Best Response Functions

Nash equilibria are mutual best responses



Applying game theory to a “real” strategy case

Our case: Deregulation of Spain’s passenger rail industry





You run strategy at the national rail incumbent. In two years, rivals can run trains on your most profitable routes.

A **Fight**

Cut fares, add frequencies, Defend every passenger

B **Tolerate**

Let them in, protect your margins

C **It depends…**

…on what exactly?

Quick poll: raise your hand. Then defend your choice in one sentence.

Using game theory to make management decisions

Textbook game theory

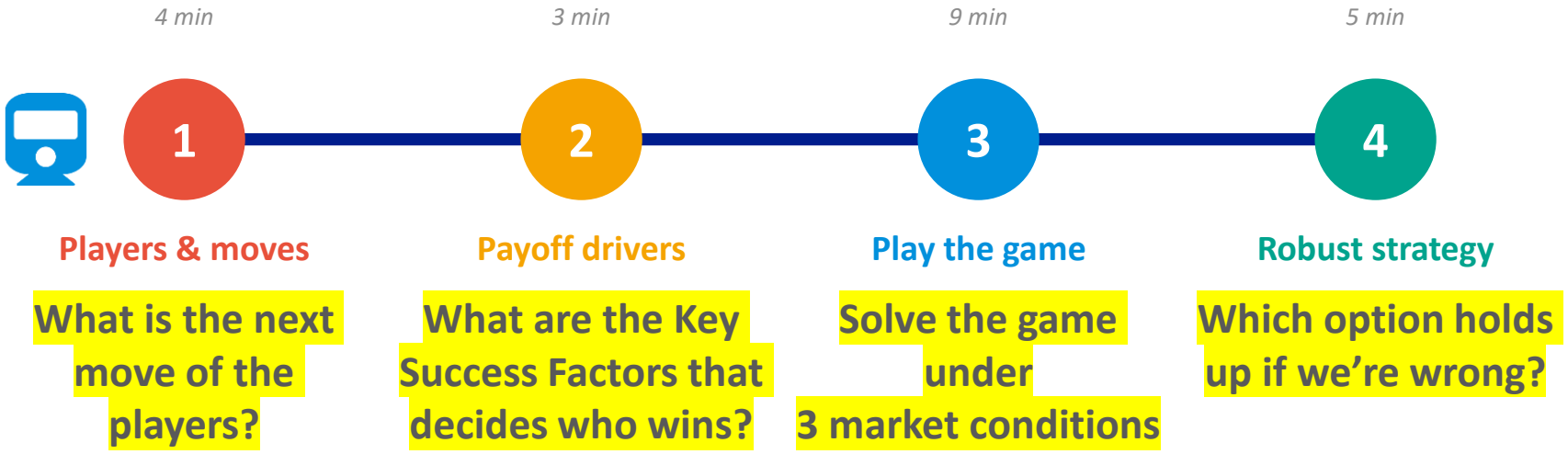
- One precisely defined game (think prisoner's dilemma)
- Solve for the equilibrium
- Output: a single "right" answer
- Fragile: tweak an assumption and the answer flips

Game theory as a decision tool

- A short, manageable list of moves for each player
- Play a sequence of games under different conditions
- Output: a map of likely outcomes
- Pick the best robust option and know its break points

Based on Lindstädt & Müller (2010), McKinsey Quarterly

We apply game theory to a real case



1 Players & moves

What is the next move of the players?

2 Payoff drivers

What are the Key Success Factors that decides who wins?

3 Play the game

Solve the game under 3 market conditions

4 Robust strategy

Which option holds up if we're wrong?

Your turn: list the moves of the Players

A low-cost train company wants to run long-distance trains on the incumbent's routes. What can each side do?

The entrant (attacker) could…

1. _____
2. _____
3. _____
4. _____

The incumbent could respond by…

1. _____
2. _____
3. _____
4. _____

The four main moves of each side

Entrant		Incumbent	
Imitate	Similar or identical service	**Ignore**	Don't react at all
Attack	More attractive offer: cheaper or more frequent	**Counterattack**	Contest via price, frequency, schedules
Specialize	Niche, e.g. peak hours only	**Coexist**	Cede some routes, learn to share
Differentiate	Distinct offer, e.g. low-cost for leisure travelers	**Exit**	Stop serving the route



Players & moves
What is the next move of the players?

Payoff drivers
What are the Key Success Factors that decides who wins?

Play the game
Solve the game under 3 market conditions

Robust strategy
Which option holds up if we're wrong?

What decides who wins? → Key Success Factors

The same move can be brilliant or ruinous. Name the market conditions that change the payoffs.

- If a new train appears, where do its passengers come from?
- Who can run a train more cheaply — and why?
- Why might a passenger stay with the incumbent even if it is pricier?
- How big a fare gap does it take to make people switch?

Four drivers set the payoffs

Demand growth

Does entry grow the market (travelers leave cars and planes) or split a fixed pie?

Cost differences

Entrants run leaner; incumbents have scale but also complexity and public-service duties

Network advantage

Incumbents connect everywhere; feeder routes and seamless transfers keep passengers loyal

Price sensitivity

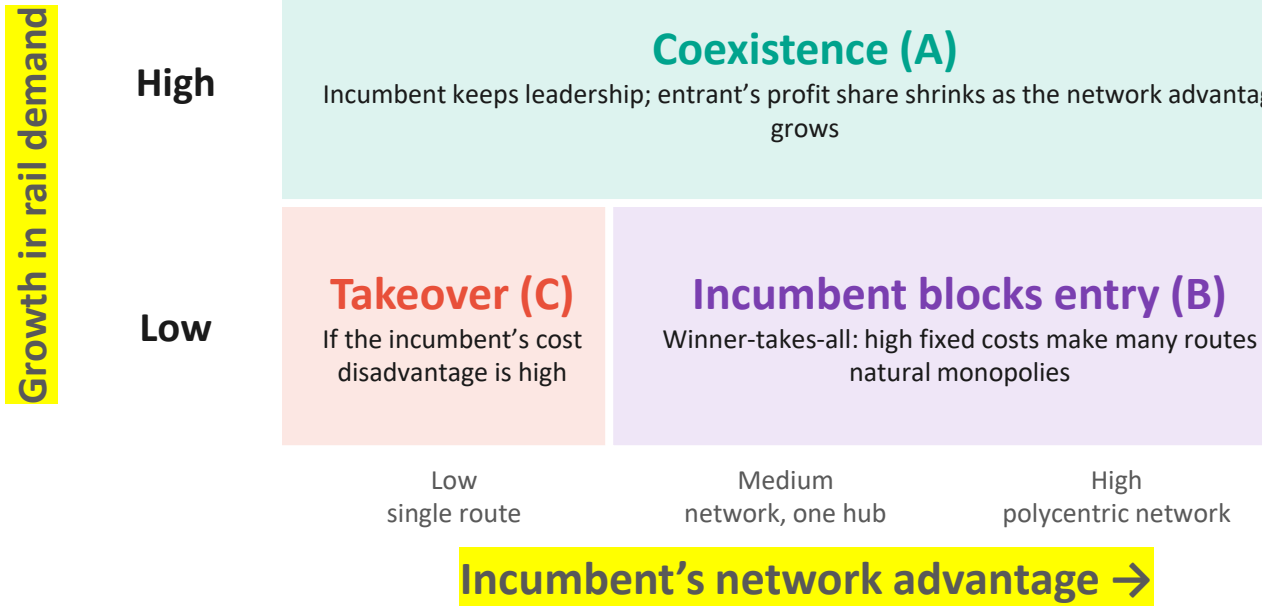
How large a fare gap do passengers need before they switch trains, or switch to rail?

1 **Players & moves** — What is the next move of the players?

2 **Payoff drivers** — What are the Key Success Factors that decides who wins?

3 **Play the game** — Solve the game under 3 market conditions

4 **Robust strategy** — Which option holds up if we're wrong?



Discuss

Which option does OK in every cell? That is the best robust option.

Break point: how weak must demand growth be before "tolerate" should become "fight"?

Simplified from Lindstädt & Müller (2010), Exhibit 1

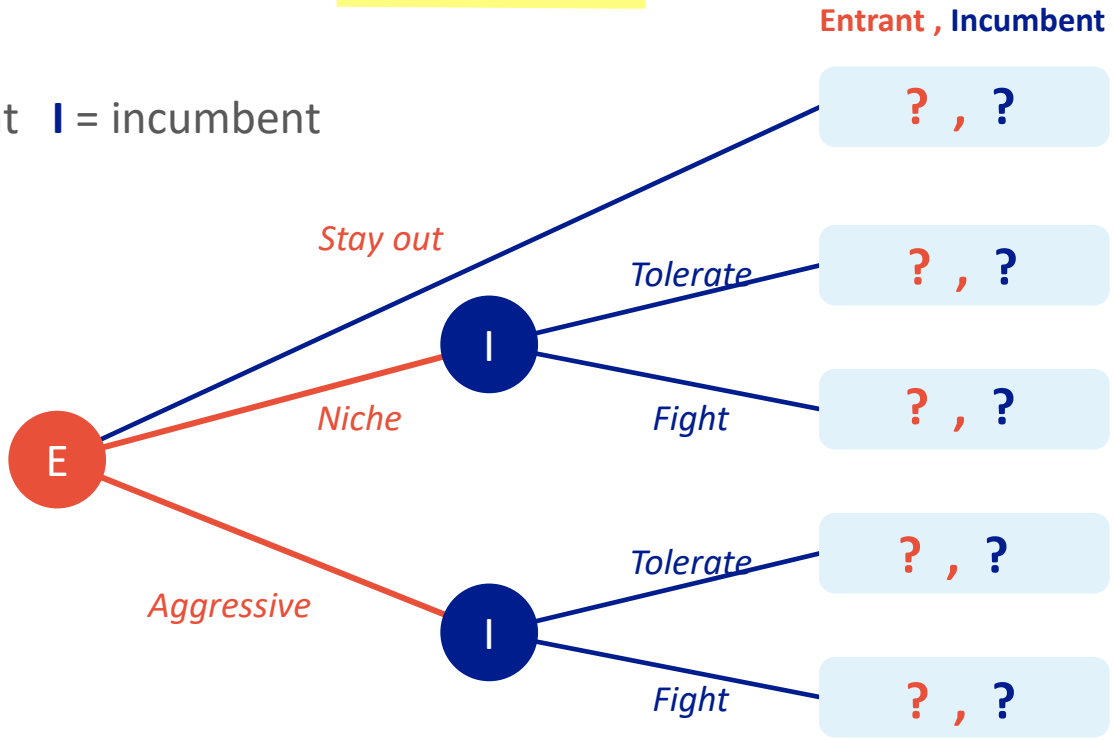
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E = entrant **I** = incumbent



How to solve it

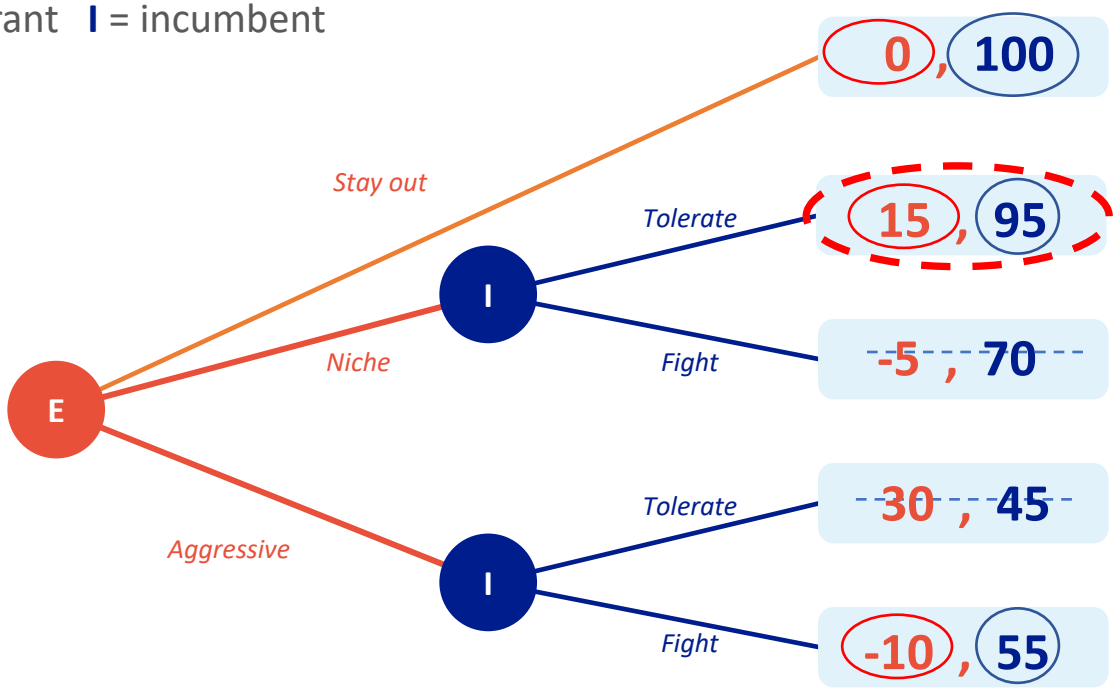
1. Payoffs = annual route profit, €m
2. Start at the end: what would the incumbent do after each entry move?
3. Anticipating that, what does the entrant choose?
4. Name the outcome in business terms

Groups A, B, C each get a different market. Same moves — different payoffs.

Group A: Growing market

Let´s share the pie

E = entrant I = incumbent



Market conditions

Demand growth

High: new services pull travelers from planes and cars

Cost gap

Incumbent's costs clearly higher

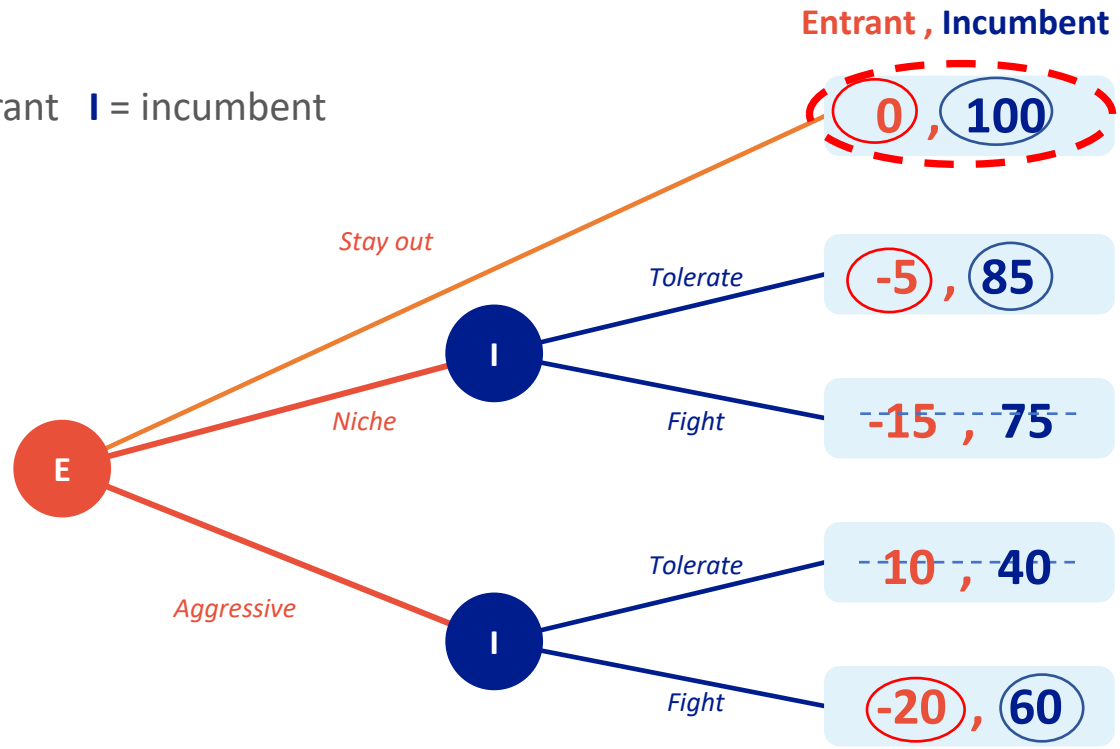
Network

Incumbent runs a large network

Solve by backward induction: what happens, and what should the incumbent expect?

Group B: Incumbent has a Strong network

Strong Differentiation Advantage



Market conditions

Demand growth

Low: rivals just split existing passengers

Cost gap

Modest

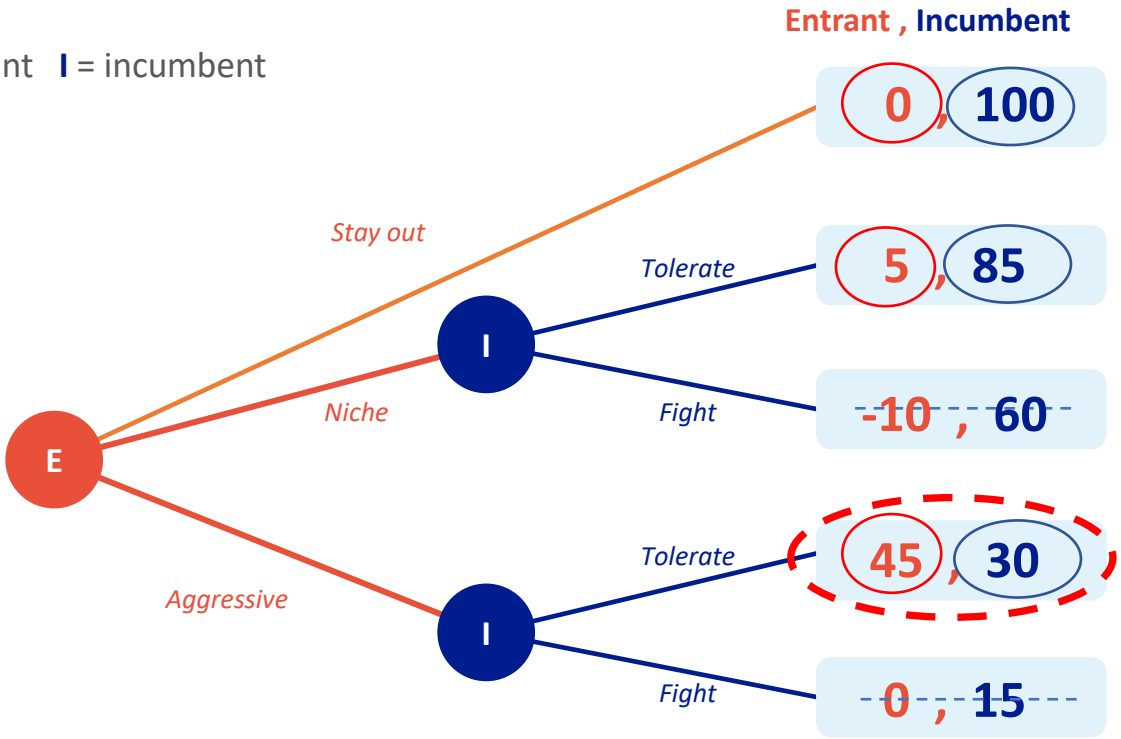
Network

Strong: many connections

Solve by backward induction: what happens, and what should the incumbent expect?

Group C: Weak incumbent

Price War



Market conditions

Demand growth

Low: rivals just split existing passengers

Cost gap

Large: legacy costs, public-service duties

Network

Weak: essentially a single route

Solve by backward induction: what happens, and what should the incumbent expect?

Same moves, three different worlds

Coexistence

A **Growing market**

Coexistence

Incumbent tolerates a niche, fights an attack → entrant goes niche

Payoffs (I , E)
15 , 95

Entrant stays out

B **Fixed pie, strong network**

Entry blocked

Fighting beats tolerating → the threat is credible

Payoffs (I , E)
0 , 100

Entrant attacks and wins

C **Fixed pie, weak incumbent**

Takeover

A price war hurts the incumbent most → threat not credible

Payoffs (I , E)
45 , 30

The incumbent's best response decision:

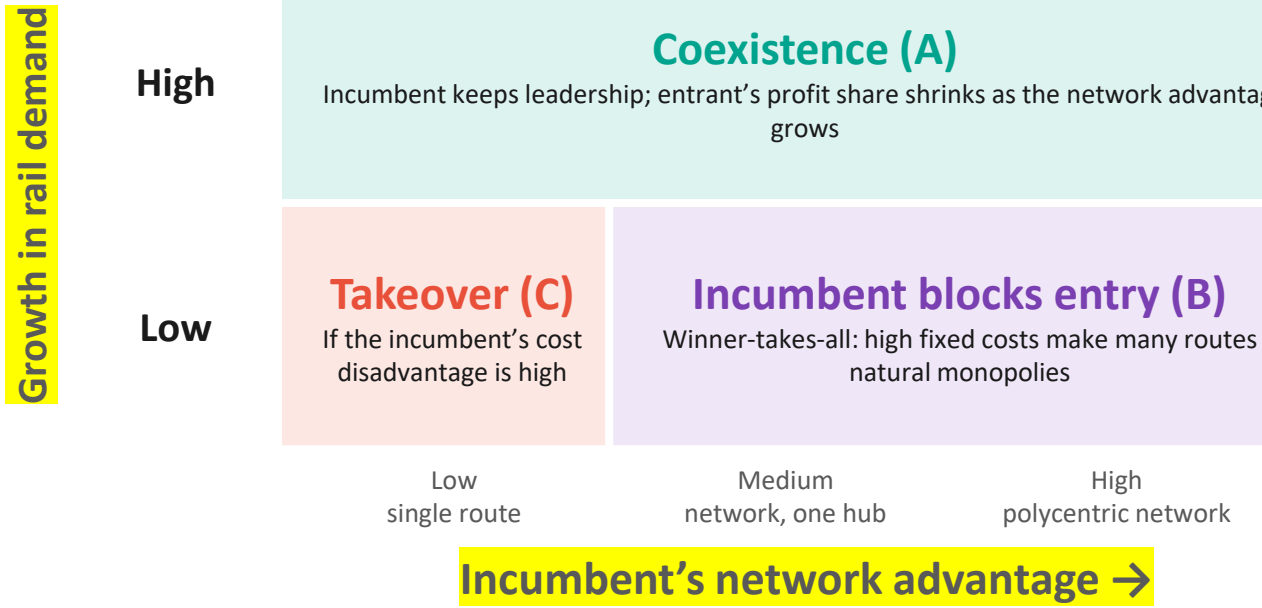
whether fighting is cheaper than tolerating — and the entrant knows it.

1 **Players & moves** — What is the next move of the players?

2 **Payoff drivers** — What are the Key Success Factors that decides who wins?

3 **Play the game** — Solve the game under 3 market conditions

4 **Robust strategy** — Which option holds up if we're wrong?



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Which option does OK in every cell? That is the best robust option.

Break point: how weak must demand growth be before "tolerate" should become "fight"?

Simplified from Lindstädt & Müller (2010), Exhibit 1

Reality check: Madrid–Barcelona

Demand grew strongly, as in Scenario A, yet the market ended in a price war.

- **2019** — Renfe's AVE is the only operator on Spain's high-speed lines
- **May 2021** — Ouigo (SNCF) enters with a low-cost, high-capacity service
- **Jun 2021** — Renfe launches Avlo, its own low-cost brand
- **Nov 2022** — Iryo (Trenitalia-led) enters with a premium-to-budget offer
- **2025** — Record 44.5m high-speed passengers, nearly 2x 2019; fares on the corridor down by a quarter or more
- **Sep 2025** — Renfe pulls Avlo from Madrid–Barcelona; average fares up ~40% year on year

NO operator is making a profit

…on most routes.

Madrid–Barcelona only reached a slim ~6% margin in 2026.

Sources: Railway Gazette (Jul 2026); Ara (2025); company announcements

Thanks a lot!

Any questions?

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Optional: https://www.youtube.com/watch?v=S49oKtISkqI

